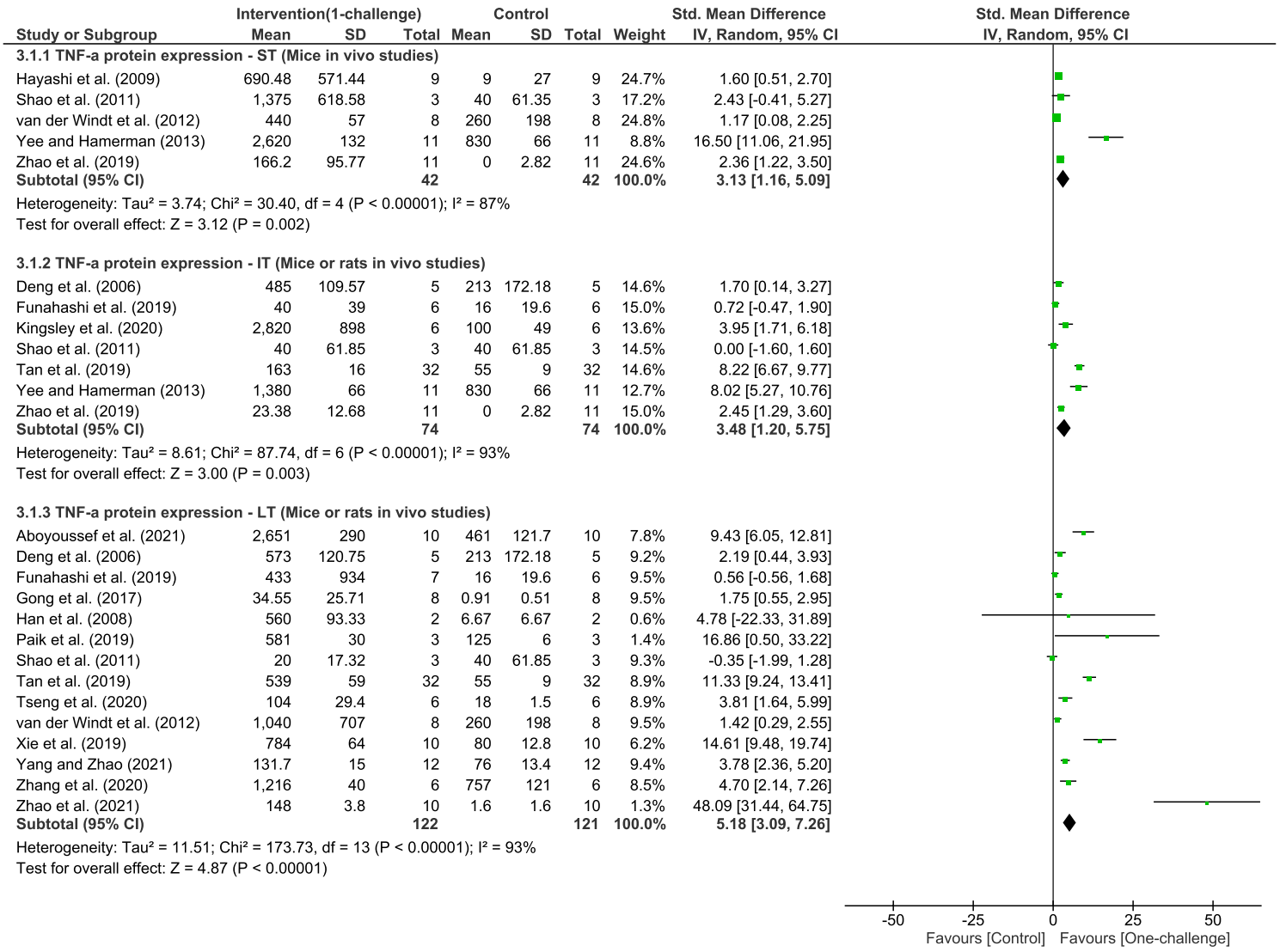

Intervention(1-challenge)
Control
Std. Mean Difference
Std. Mean Difference
Study or Subgroup
Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
3.1.1 TNF-a protein expression - ST (Mice in vivo studies)
Hayashi et al. (2009) 690.48 571.44 9 9 27 9 24.7% 1.60 [0.51, 2.70]
Shao et al. (2011) 1,375 618.58 3 40 61.35 3 17.2% 2.43 [-0.41, 5.27]
van der Windt et al. (2012) 440 57 8 260 198 8 24.8% 1.17 [0.08, 2.25]
Yee and Hamerman (2013) 2,620 132 11 830 66 11 8.8% 16.50 [11.06, 21.95]
Zhao et al. (2019) 166.2 95.77 11 0 2.82 11 24.6% 2.36 [1.22, 3.50]
Subtotal (95% CI) 42 42 100.0% 3.13 [1.16, 5.09]
Heterogeneity: Tau² = 3.74; Chi² = 30.40, df = 4 (P < 0.00001); I² = 87%
Test for overall effect: Z = 3.12 (P = 0.002)
3.1.2 TNF-a protein expression - IT (Mice or rats in vivo studies)
Deng et al. (2006) 485 109.57 5 213 172.18 5 14.6% 1.70 [0.14, 3.27]
Funahashi et al. (2019) 40 39 6 16 19.6 6 15.0% 0.72 [-0.47, 1.90]
Kingsley et al. (2020) 2,820 898 6 100 49 6 13.6% 3.95 [1.71, 6.18]
Shao et al. (2011) 40 61.85 3 40 61.85 3 14.5% 0.00 [-1.60, 1.60]
Tan et al. (2019) 163 16 32 55 9 32 14.6% 8.22 [6.67, 9.77]
Yee and Hamerman (2013) 1,380 66 11 830 66 11 12.7% 8.02 [5.27, 10.76]
Zhao et al. (2019) 23.38 12.68 11 0 2.82 11 15.0% 2.45 [1.29, 3.60]
Subtotal (95% CI) 74 74 100.0% 3.48 [1.20, 5.75]
Heterogeneity: Tau² = 8.61; Chi² = 87.74, df = 6 (P < 0.00001); I² = 93%
Test for overall effect: Z = 3.00 (P = 0.003)
3.1.3 TNF-a protein expression - LT (Mice or rats in vivo studies)
Aboyoussef et al. (2021) 2,651 290 10 461 121.7 10 7.8% 9.43 [6.05, 12.81]
Deng et al. (2006) 573 120.75 5 213 172.18 5 9.2% 2.19 [0.44, 3.93]
Funahashi et al. (2019) 433 934 7 16 19.6 6 9.5% 0.56 [-0.56, 1.68]
Gong et al. (2017) 34.55 25.71 8 0.91 0.51 8 9.5% 1.75 [0.55, 2.95]
Han et al. (2008) 560 93.33 2 6.67 6.67 2 0.6% 4.78 [-22.33, 31.89]
Paik et al. (2019) 581 30 3 125 6 3 1.4% 16.86 [0.50, 33.22]
Shao et al. (2011) 20 17.32 3 40 61.85 3 9.3% -0.35 [-1.99, 1.28]
Tan et al. (2019) 539 59 32 55 9 32 8.9% 11.33 [9.24, 13.41]
Tseng et al. (2020) 104 29.4 6 18 1.5 6 8.9% 3.81 [1.64, 5.99]
van der Windt et al. (2012) 1,040 707 8 260 198 8 9.5% 1.42 [0.29, 2.55]
Xie et al. (2019) 784 64 10 80 12.8 10 6.2% 14.61 [9.48, 19.74]
Yang and Zhao (2021) 131.7 15 12 76 13.4 12 9.4% 3.78 [2.36, 5.20]
Zhang et al. (2020) 1,216 40 6 757 121 6 8.5% 4.70 [2.14, 7.26]
Zhao et al. (2021) 148 3.8 10 1.6 1.6 10 1.3% 48.09 [31.44, 64.75]
Subtotal (95% CI) 122 121 100.0% 5.18 [3.09, 7.26]
Heterogeneity: Tau² = 11.51; Chi² = 173.73, df = 13 (P < 0.00001); I² = 93%
Test for overall effect: Z = 4.87 (P < 0.00001)
-50
-25
0
25
50
Favours [Control]
Favours [One-challenge]